

Phase 4: Project Planning Phase

This phase outlines the Agile development methodology, establishing the product backlog, user stories with acceptance criteria, sprint schedules, and project velocity metrics.

1. Product Backlog & Sprint Schedule

Development was executed across three main sprints, targeting key feature increments:

Sprint	Epic	USN ID	User Story / Task	Story Points	Priority	Status
Sprint 1	Foundation & Auth	USN-001	Project setup: Express API server, React frontend compilation structure, MongoDB initial model bindings.	3	High	Completed
Sprint 1	Foundation & Auth	USN-002	Secure user registration with Bcrypt hashing and validation checks.	2	High	Completed
Sprint 1	Foundation & Auth	USN-003	User login session initialization returning signed JWT tokens in HTTP-Only cookies.	2	High	Completed

Sprint 2	Discovery & Cart	USN-004	Product catalog search with live autocomplete suggestions and category filters.	3	High	Completed
Sprint 2	Discovery & Cart	USN-005	Persistent shopping cart context preserving products, totals, and quantities on page reload.	3	High	Completed
Sprint 2	Discovery & Cart	USN-006	Product wishlist management: saving favorites and supporting "One-Click Move to Cart".	2	Medium	Completed
Sprint 2	Checkout & Fulfill	USN-007	Multi-step checkout wizard integrating active coupon rules validation and tax estimations.	3	High	Completed
Sprint 3	Checkout & Fulfill	USN-008	Stripe mock integration to generate payment intents and verify checkouts.	3	High	Completed

Sprint 3	Feedback & Logs	USN-009	Order history lookup and visual timeline fulfillment tracking.	2	Medium	Completed
Sprint 3	Feedback & Logs	USN-010	Verified buyer reviews and helpful feedback upvote system.	2	Medium	Completed
Sprint 3	Administration	USN-011	Unified Admin Dashboard panel with sales growth charts, inventory tables, and coupon configuration controls.	5	High	Completed

2. Detailed User Stories & Acceptance Criteria

USN-002: User Registration

- **User Story:** As a new customer, I can register a profile by entering my name, email, and password, so that I can shop and track orders.
- **Acceptance Criteria:**
 - Must block submissions with duplicate emails or weak passwords (< 6 characters).
 - Password must be securely encrypted using Bcrypt before saving to MongoDB.
 - Successful registration redirects the user directly to the home screen with an active session.

USN-004: Autocomplete Product Search

- **User Story:** As a shopper, I can input search words and select categories, so that I can find matching electronics or clothing immediately.

- **Acceptance Criteria:**
- Submitting text queries instantly queries the [/api/products/suggestions](#) API.
- Results must respect category selections, price ranges, and minimum rating stars filters.
- Sorting options (price, rating, date) must re-order results immediately on-screen.

USN-007: Coupon Code Validation

- **User Story:** As a customer, I can apply coupon codes during checkout, so that I receive active promotional discounts on my orders.
- **Acceptance Criteria:**
- Coupon validation endpoint [/api/coupons/validate](#) checks active flags and expiry dates.
- Applied discounts must be itemized (fixed vs percentage) and deducted from the subtotal.
- Taxes (e.g. 15% VAT) and shipping speeds (e.g., Express vs Standard) must recalculate automatically.

USN-011: Unified Admin Dashboard

- **User Story:** As a store admin, I can view visual monthly sales graphs and inventory tables, so that I can manage my shop operations.
- **Acceptance Criteria:**
- Metrics charts must compile monthly revenue, category shares, and new user enrollments.
- Product inventory tables must render alerts for items dropping below set stock thresholds.
- Admin can instantly block/unblock customer logins and toggle coupon statuses.

3. Sprint Metrics & Velocity Tracker

- **Development Methodology:** Scrum framework.
- **Sprint Duration:** 2-week iterations.
- **Planned Team Velocity:** 10 Story Points per sprint.

Sprint Progress Summary





- **Sprint 1 (Weeks 1-2):** Planned: 7 SP | Completed: 7 SP
- *Focus:* Base Express structure, database models, user registry, and JWT sessions.
- **Sprint 2 (Weeks 3-4):** Planned: 11 SP | Completed: 11 SP
- *Focus:* Product catalogs, search suggestions, persistent cart context, address books, and coupon validation logic.
- **Sprint 3 (Weeks 5-6):** Planned: 12 SP | Completed: 12 SP
- *Focus:* Stripe Mock payments gateway, visual order tracking timeline, buyer review scores, and Admin sales dashboard analytics.
- **Average Velocity achieved:** 10 SP per Sprint.